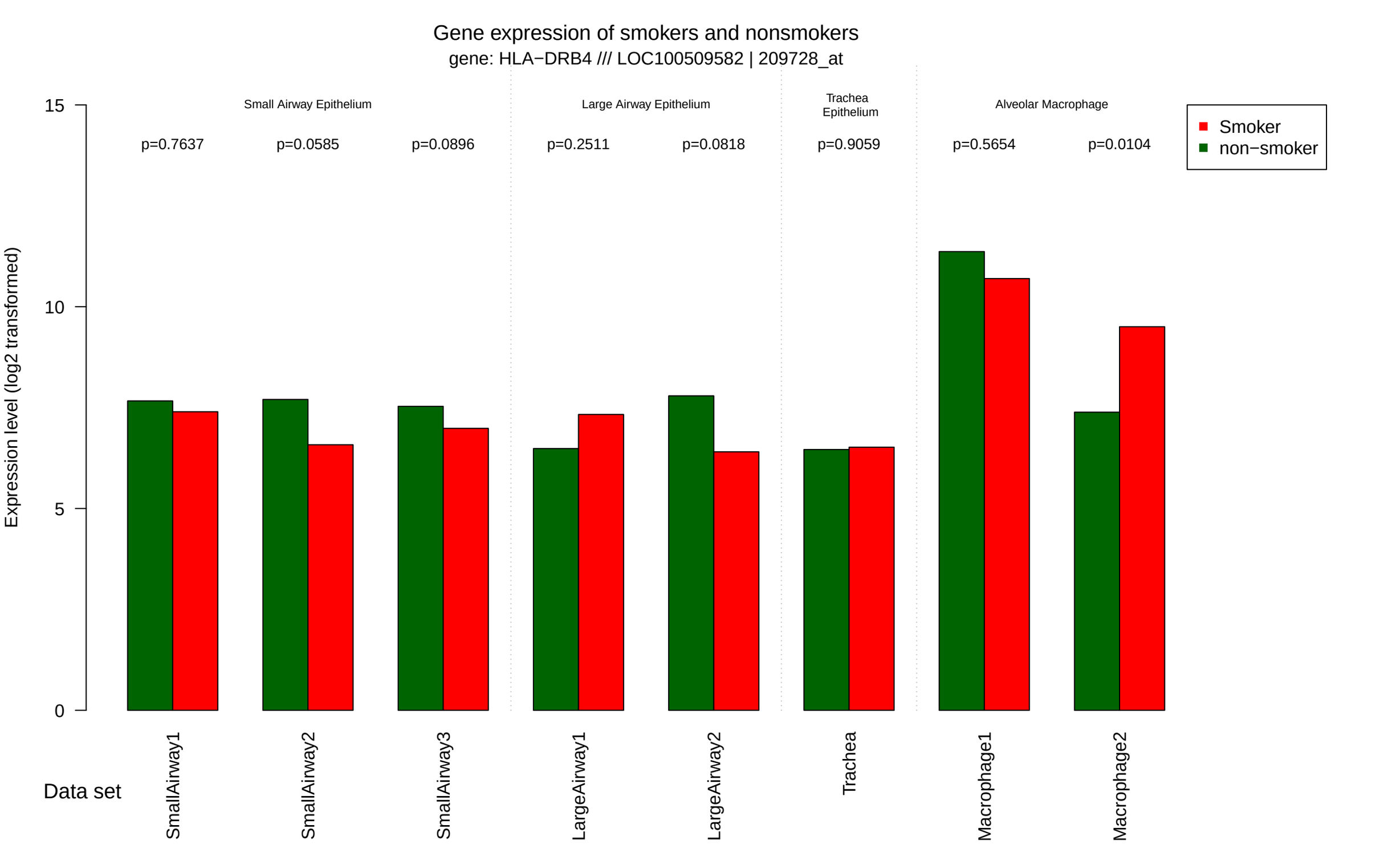
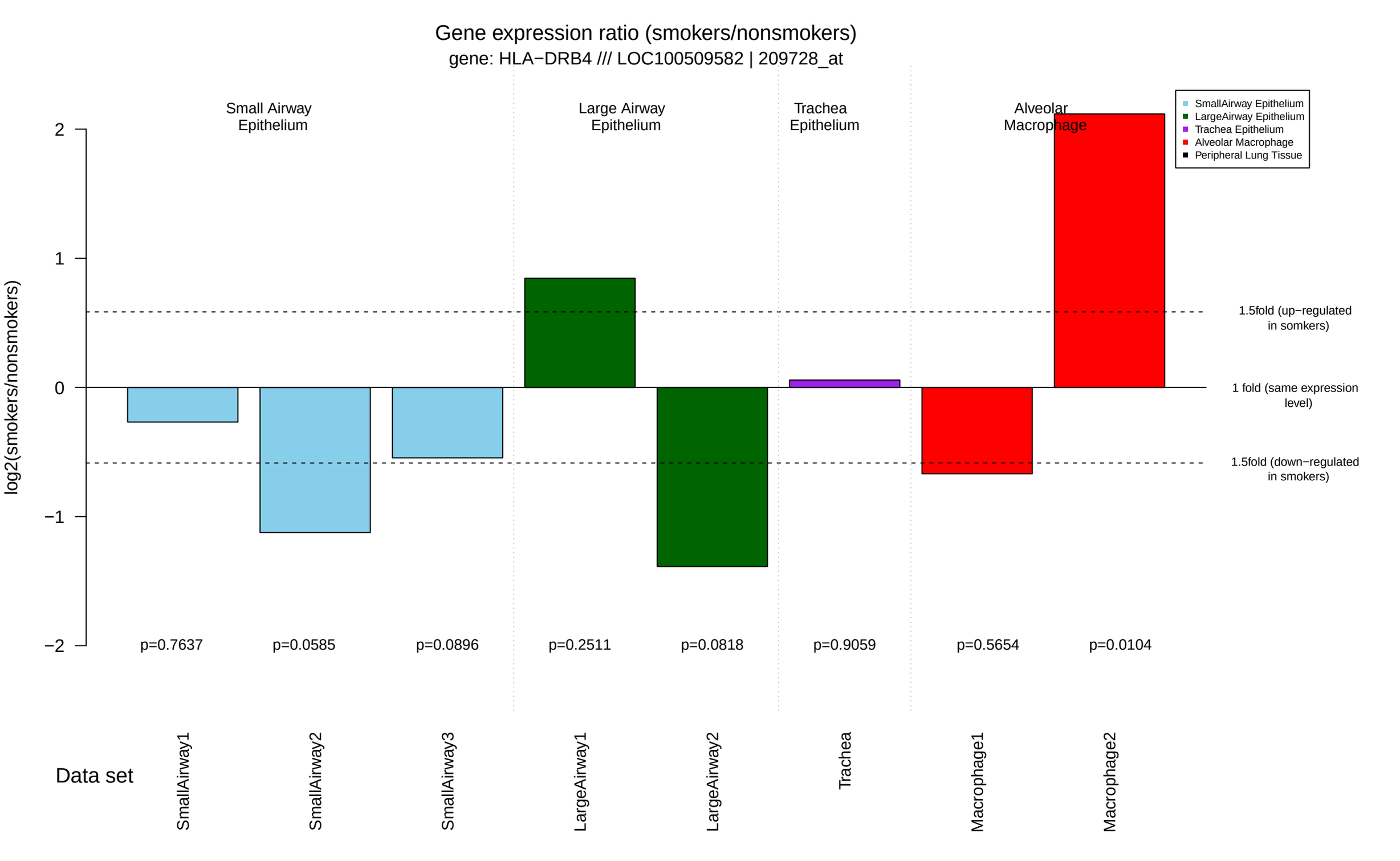
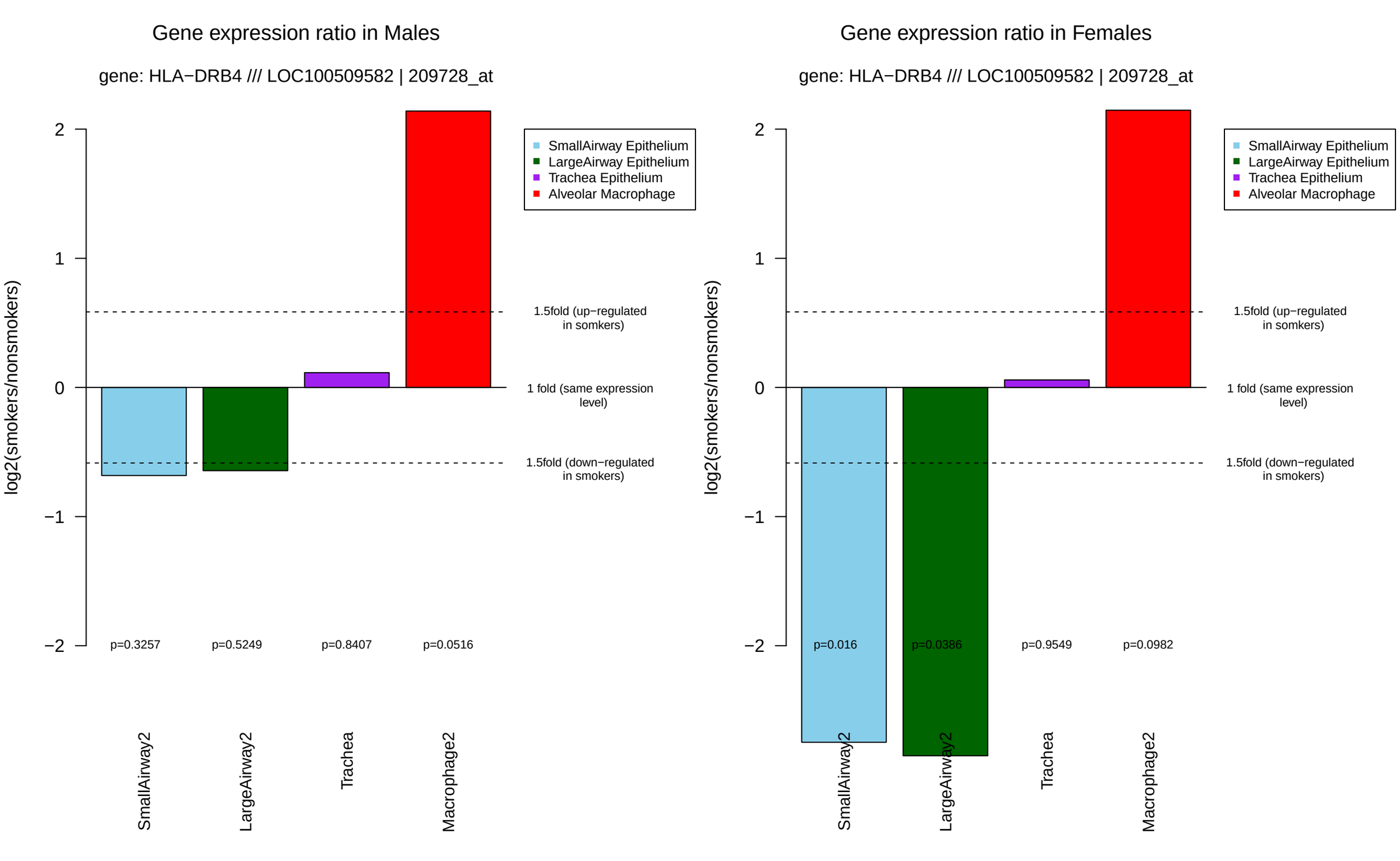
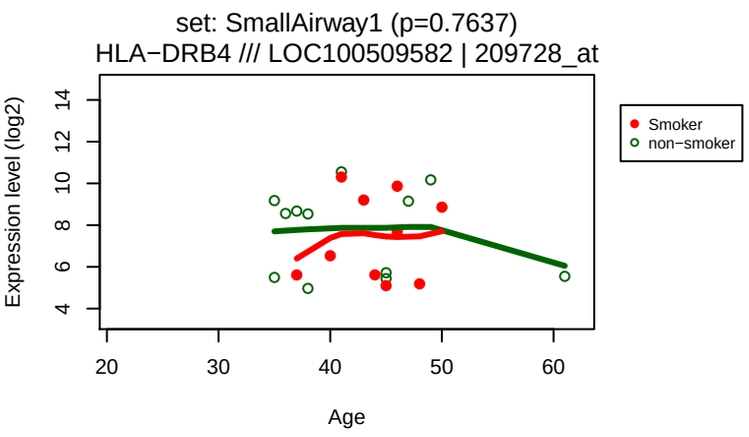
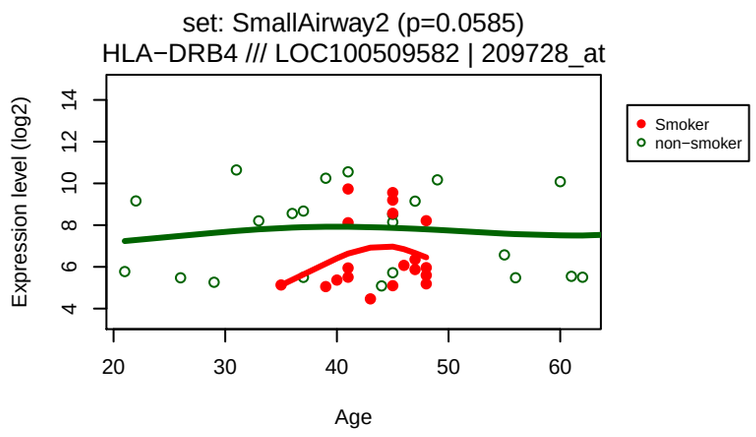
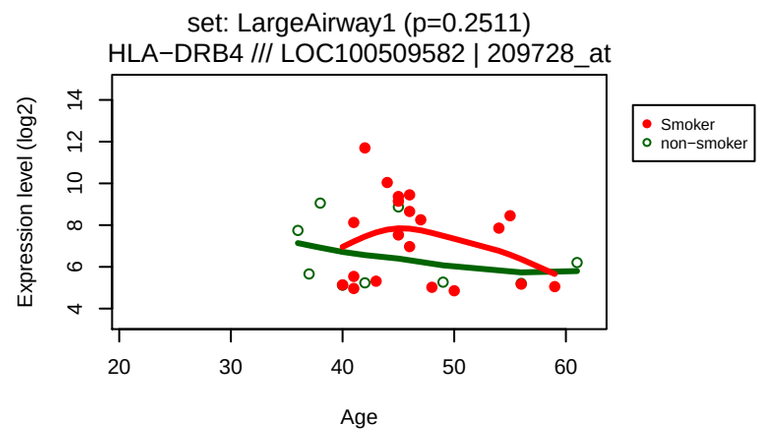

Gene expression of smokers and nonsmokers
gene: HLA−DRB4 /// LOC100509582 | 209728_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7637
p=0.0585
p=0.0896
p=0.2511
p=0.0818
p=0.9059
p=0.5654
p=0.0104
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

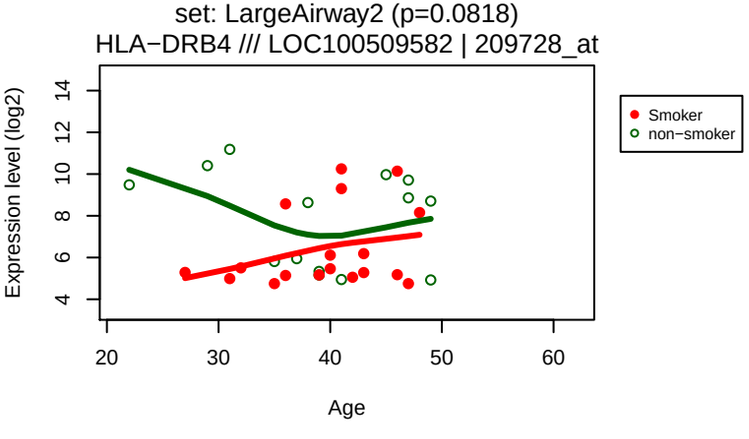

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DRB4 /// LOC100509582 | 209728_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7637
p=0.0585
p=0.0896
p=0.2511
p=0.0818
p=0.9059
p=0.5654
p=0.0104
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

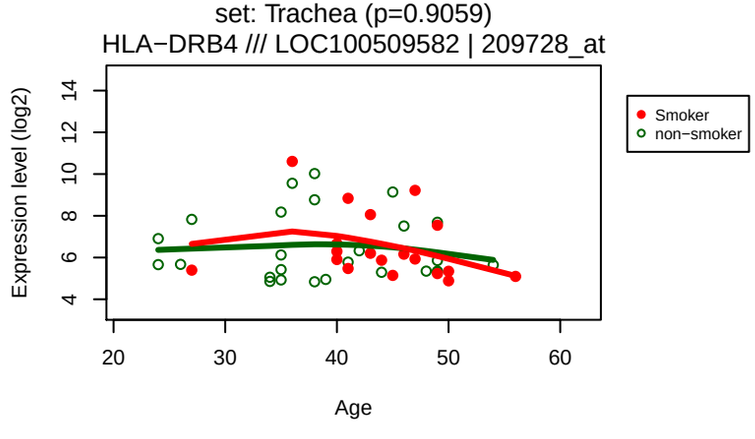

Gene expression ratio in Males
gene: HLA−DRB4 /// LOC100509582 | 209728_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3257
p=0.5249
p=0.8407
p=0.0516
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DRB4 /// LOC100509582 | 209728_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.016
p=0.0386
p=0.9549
p=0.0982
SmallAirway2
LargeAirway2
Trachea
Macrophage2

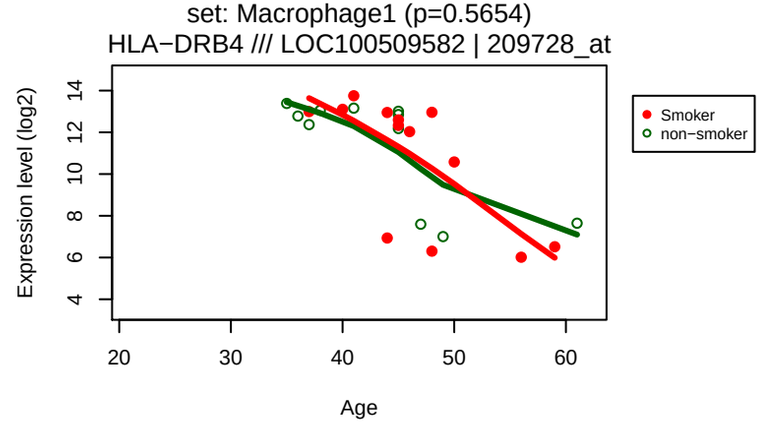

set: SmallAirway1 (p=0.7637)
HLA−DRB4 /// LOC100509582 | 209728_at
Expression level (log2)
Age
Smoker
non−smoker

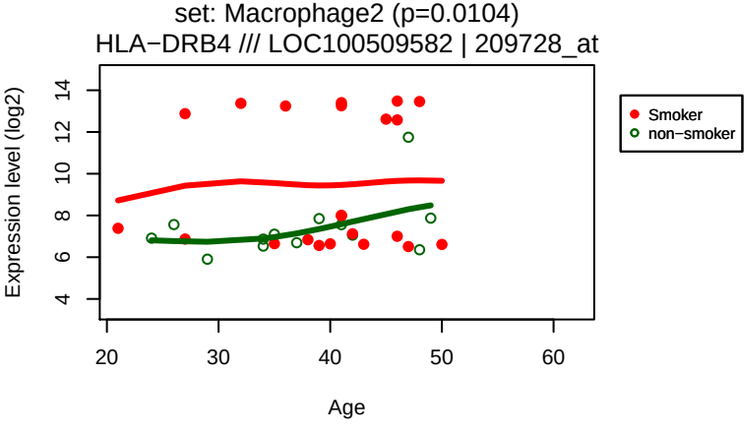

set: SmallAirway2 (p=0.0585)
HLA−DRB4 /// LOC100509582 | 209728_at
Expression level (log2)
Age
Smoker
non−smoker

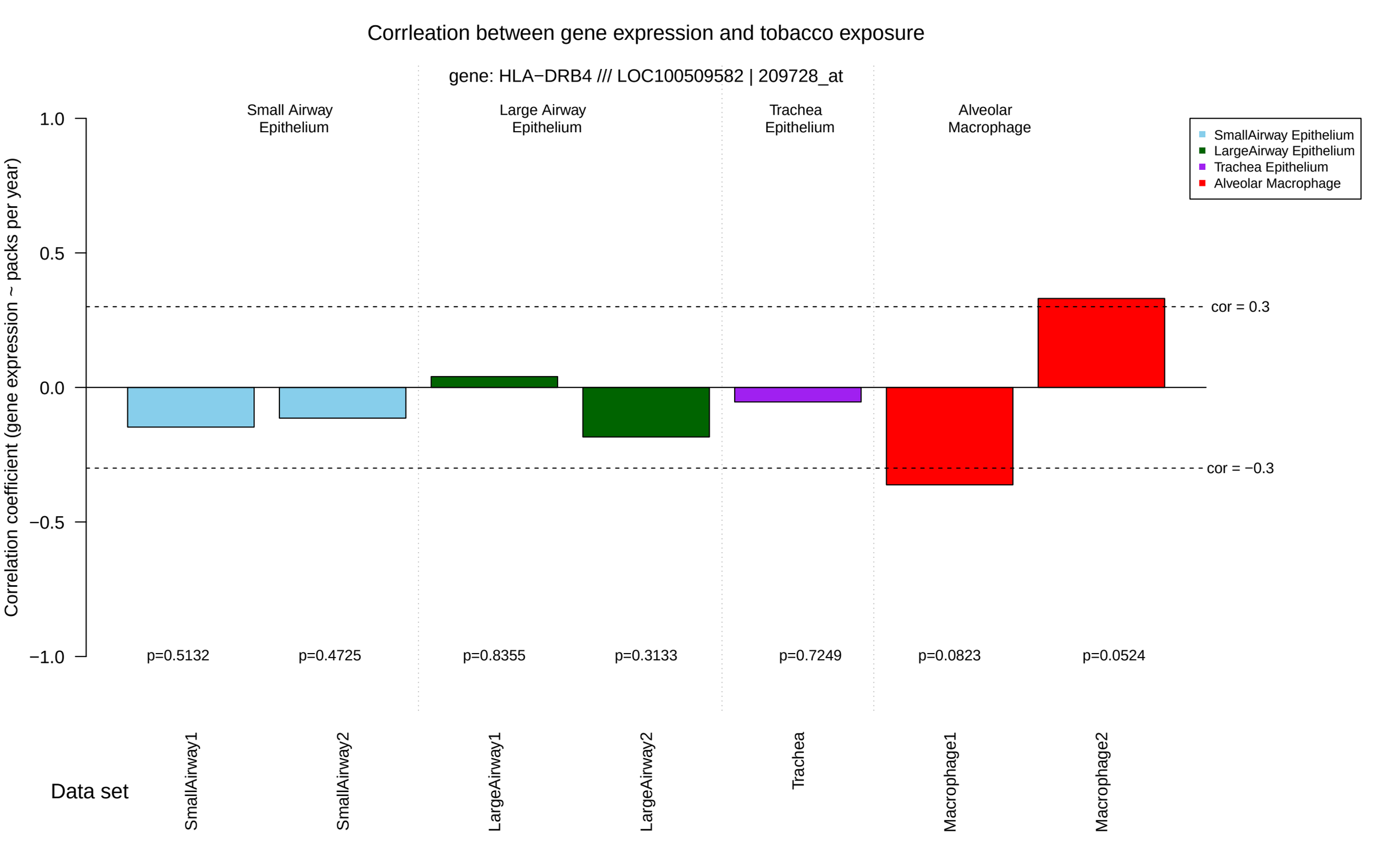

set: LargeAirway1 (p=0.2511)
HLA−DRB4 /// LOC100509582 | 209728_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0818)
HLA−DRB4 /// LOC100509582 | 209728_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9059)
HLA−DRB4 /// LOC100509582 | 209728_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5654)
HLA−DRB4 /// LOC100509582 | 209728_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0104)
HLA−DRB4 /// LOC100509582 | 209728_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DRB4 /// LOC100509582 | 209728_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5132
p=0.4725
p=0.8355
p=0.3133
p=0.7249
p=0.0823
p=0.0524
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2